

 rinna / **japanese-gpt-neox-3.6b** 

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 Text Generation  Transformers  PyTorch  Safetensors  cc100

 wikipedia  mc4  Japanese gpt_neox lm nlp

 text-generation-inference  arxiv:2404.01657  License: mit

  Train  Deploy  Use this model

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13,416



 Safetensors 

Model size 3.76B params

Tensor type F32 · F16 · U8

 Files info

 Inference Providers **NEW**

 Text Generation

This model isn't deployed by any Inference Provider.

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 Model tree for rinna/japanese-gpt-neox-3.6b

	Adapters	5 models
	Finetunes	5 models
	Merges	1 model
	Quantizations	1 model

 Datasets used to train rinna/japanese-gpt-neox-3.6b

 Spaces using rinna/japanese-gpt-neox-3.6b 6

 Collections including rinna/japanese-gpt-neox-3.6b

japanese-gpt-neox-3.6b Collection

The japanese-gpt-neox-3.6b series are pr... • 5 items • Updated Aug 26 • Δ 2

language model Collection

16 items • Updated Aug 26 • Δ 3

[japanese-gpt-neox-3.6b](#)



[Overview](#)

This repository provides a Japanese GPT-NeoX model of 3.6 billion parameters.

- **Library**

The model was trained using code based on [EleutherAI/gpt-neox](#).

- **Model architecture**

A 36-layer, 2816-hidden-size transformer-based language model.

- **Pre-training**

The model was trained on around **312.5B** tokens from Japanese CC-100, Japanese C4, and Japanese Wikipedia to optimize a traditional language modelling objective.

A final validation perplexity of **8.68** has been reached.

- **Model Series**

Variant	Link
3.6B PPO	https://huggingface.co/rinna/japanese-gpt-neox-3.6b-instruction-ppo
3.6B SFT-v2	https://huggingface.co/rinna/japanese-gpt-neox-3.6b-instruction-sft-v2
3.6B SFT	https://huggingface.co/rinna/japanese-gpt-neox-3.6b-instruction-sft
3.6B pretrained	https://huggingface.co/rinna/japanese-gpt-neox-3.6b

- **Contributors**

Tianyu Zhao and Kei Sawada

- **Release date**

March 17, 2023

[🔗](#) How to use the model

```
import torch
from transformers import AutoTokenizer, AutoModelForCausalLM

tokenizer = AutoTokenizer.from_pretrained("rinna/japanese-gpt-neox-3.6b", use_fast
model = AutoModelForCausalLM.from_pretrained("rinna/japanese-gpt-neox-3.6b")

if torch.cuda.is_available():
    model = model.to("cuda")

text = "西田幾多郎は、"
token_ids = tokenizer.encode(text, add_special_tokens=False, return_tensors="pt")
```

```

with torch.no_grad():
    output_ids = model.generate(
        token_ids.to(model.device),
        max_new_tokens=100,
        min_new_tokens=100,
        do_sample=True,
        temperature=0.8,
        pad_token_id=tokenizer.pad_token_id,
        bos_token_id=tokenizer.bos_token_id,
        eos_token_id=tokenizer.eos_token_id
    )

output = tokenizer.decode(output_ids.tolist()[0])
print(output)
"""西田幾多郎は、この「絶対矛盾的自己同一」を「世界の自己同一」と置きかえ、さらに西田哲学を出発点

```

🔗 Tokenization

The model uses a sentencepiece-based tokenizer.

- The tokenizer has a vocabulary size of 32,000.
- It uses sentencepiece's byte fallback feature to decompose unknown text pieces into UTF-8 byte pieces and to avoid producing `<UNK>` tokens.
- sentencepiece's `--add_dummy_prefix` option was turned off so that a leading whitespace will not be prepended automatically.

```

print(tokenizer.tokenize("吾輩は猫である"))
# ['吾', '輩', 'は', '猫', 'である']
# instead of ['_', '吾', '輩', 'は', '猫', 'である'] as in rinna/japanese-gpt-

```

- sentencepiece's `--remove_extra_whitespaces` option was turned off so that leading, trailing, and duplicate whitespaces are reserved.

```

print(tokenizer.tokenize(" 吾輩は 猫である "))
# ['_', '_', '吾', '輩', 'は', '_', '_', '猫', 'である', '_', '_', '_']

```

```
# instead of ['_', '吾', '輩', 'は', '猫', 'である'] as in rinna/japanese-gpt
```

- Don't forget to set `use_fast=False` to make the above features function correctly.

```
good_tokenizer = AutoTokenizer.from_pretrained("rinna/japanese-gpt-neox-3.6b")
bad_tokenizer = AutoTokenizer.from_pretrained("rinna/japanese-gpt-neox-3.6b")

print(good_tokenizer.decode(good_tokenizer.encode("გამარჯობა 吾輩は 猫である"))
# 'გამარჯობა 吾輩は 猫である </s>'
print(bad_tokenizer.decode(bad_tokenizer.encode("გამარჯობა 吾輩は 猫である"))
# 'გამარ[UNK]ობა 吾輩は 猫である </s>'
```

How to cite

```
@misc{rinna-japanese-gpt-neox-3.6b,
  title = {rinna/japanese-gpt-neox-3.6b},
  author = {Zhao, Tianyu and Sawada, Kei},
  url = {https://huggingface.co/rinna/japanese-gpt-neox-3.6b}
}

@inproceedings{sawada2024release,
  title = {Release of Pre-Trained Models for the {J}apanese Language},
  author = {Sawada, Kei and Zhao, Tianyu and Shing, Makoto and Mitsui, Kentaro and
    Sawada, Kei},
  booktitle = {Proceedings of the 2024 Joint International Conference on Computer
    Vision and Pattern Recognition},
  month = {5},
  year = {2024},
  pages = {13898--13905},
  url = {https://aclanthology.org/2024.lrec-main.1213},
  note = {\url{https://arxiv.org/abs/2404.01657}}
}
```

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